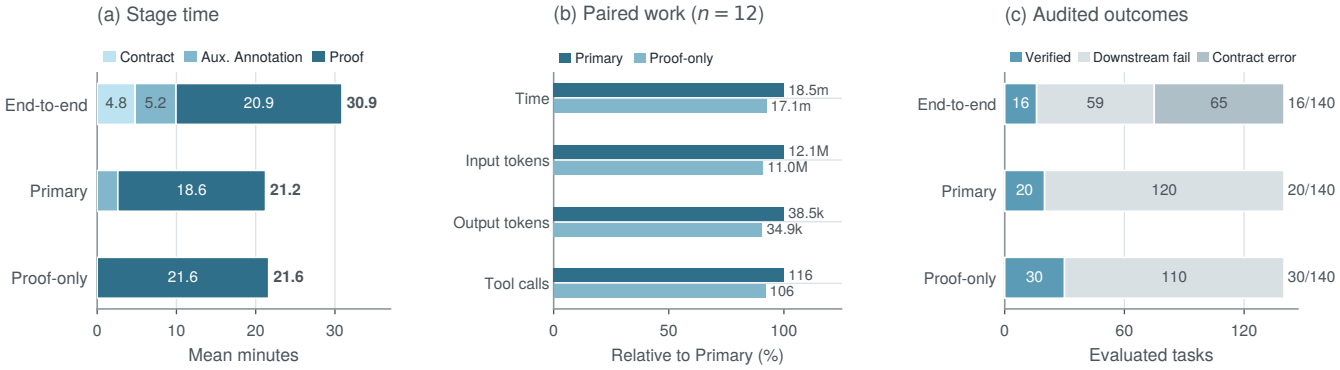
(a) Stage time
Contract
Aux. Annotation
Proof
End-to-end
Primary
Proof-only
4.8
5.2
20.9
30.9
18.6
21.2
21.6
21.6
0
10
20
30
Mean minutes
(b) Paired work (n = 12)
Primary
Proof-only
Time
Input tokens
Output tokens
Tool calls
18.5m
17.1m
12.1M
11.0M
38.5k
34.9k
116
106
0
50
100
Relative to Primary (%)
(c) Audited outcomes
Verified
Downstream fail
Contract error
End-to-end
Primary
Proof-only
16
59
65
16/140
20
120
20/140
30
110
30/140
0
60
120
Evaluated tasks